

# Spine vs. GraSU+ReGraph

## Correctness-Gated Simulator Evaluation and FPGA Feasibility

Three Graph Algorithms | E2E and Update Performance | Memory Behavior | HBM Energy | Routed Resource and Timing

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### Abstract

This evidence pack evaluates normalized Spine and GraSU+ReGraph architectures with an execution-driven simulator and routed FPGA feasibility artifacts. The reported compact-slice comparison covers weighted SSSP, full PageRank, and thresholded residual PageRank; every performance row is admitted only after architecture-precision, independent mathematical, and cross-system final-state checks pass. Claims are scoped explicitly below so that compact-slice simulation, dense-update evidence, HBM-only energy, and non-iso-functional FPGA builds are not conflated.

### Acceptance coverage

| Requirement              | Status          | Delivered evidence                                                                                                                                                                                  | Direct reference         |
|--------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Correctness              | <b>Complete</b> | All reported rows pass two numerical oracles plus graph-state and cross-system result checks.                                                                                                       | Figs. 1–11               |
| Three algorithms and E2E | <b>Complete</b> | A 540,000-edge, 157,107-vertex, three-partition AskUbuntu slice at batches 8/64/4,096, plus five 8,192-edge real-topology breadth slices.                                                           | Figs. ??, 1; Tables 1, 2 |
| Update performance       | <b>Complete</b> | Full PageRank exercises the common structural path across insert, delete, mixed, and weight-change scenarios; throughput counts semantic mutations.                                                 | Figs. 2, 3               |
| Memory behavior          | <b>Complete</b> | Three-algorithm logical traffic, locality, DRAM row behavior, burst amplification, and finite-queue pressure with request conservation.                                                             | Figs. 5, 6               |
| Dense batches            | <b>Partial</b>  | Full PageRank on three real topologies at batches 64/512/4,096; broader algorithm and multi-partition coverage remains open.                                                                        | Fig. 7                   |
| Energy                   | <b>Partial</b>  | Matched all-controller HBM energy for all three algorithms plus routed, vectorless component power attribution. Workload-calibrated on-chip energy remains open.                                    | Figs. 8, 9               |
| Area and timing          | <b>Partial</b>  | Routed U55C resources/setup timing and a 32nm SRAM-capacity area projection are available. A full standard-cell ASIC area is not claimed.                                                           | Table 3; Figs. 13, 10    |
| Simulator runtime        | <b>Partial</b>  | All compact runs complete; a correctness-gated 50,000-edge full-PageRank slice completes below the 1,800-second per-system gate.                                                                    | Fig. 11                  |
| Ablation and scalability | <b>Partial</b>  | Two bounded Spine optimizations have controlled cycle/request ablations; shared-HBM $\pm 20\%$ sensitivity has no strict ranking inversion. Replicated-kernel and full-dataset scaling remain open. | Fig. 12                  |

**Claim rule.** Every plotted performance row must pass the architecture-precision oracle, the independent mathematical oracle, and the cross-system final-state check. Compact slices and larger validation slices are reported as separate populations. The 540,000-edge AskUbuntu experiment is multi-partition but is still a slice, not a full-dataset claim.

# 1 Paper-scale multi-partition comparison

## Evidence gate not complete

Waiting for all nine 540K-edge three-algorithm pairs to pass.

| Topology                 | Vertices | Base edges | G+R partitions | Batches    |
|--------------------------|----------|------------|----------------|------------|
| AskUbuntu temporal slice | 157,107  | 540,000    | 3              | 8/64/4,096 |

Table 1: Paper-scale workload. Base edges are unique, non-self edges. Each batch uses a separate later-event insertion slice and is checked against two numerical oracles plus cross-system final state before admission.

# 2 Compact breadth validation

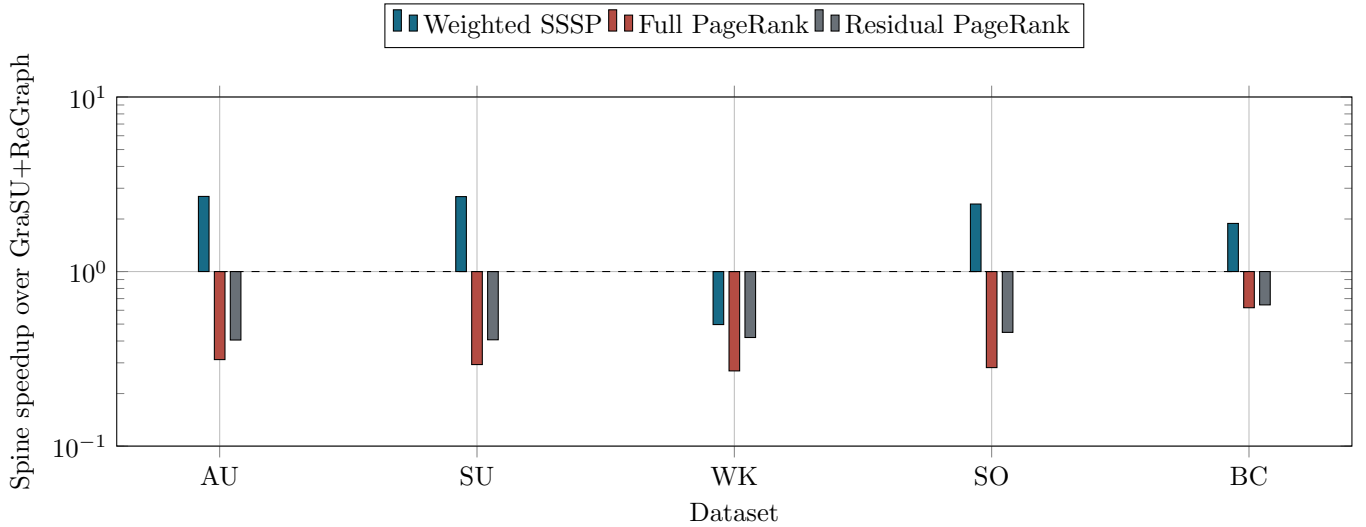
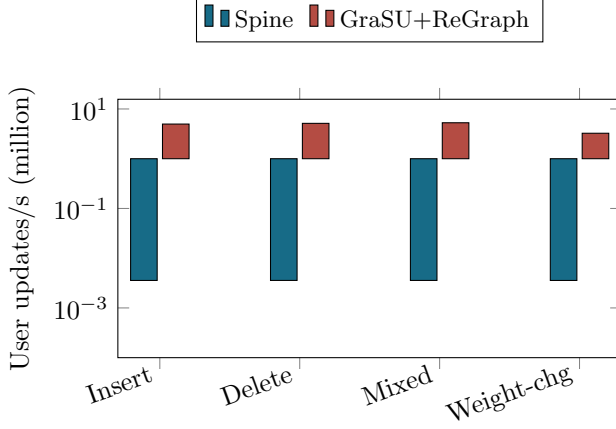


Figure 1: Insert-batch-8 E2E comparison on five compact temporal slices. Values above one favor Spine; values below one favor GraSU+ReGraph.

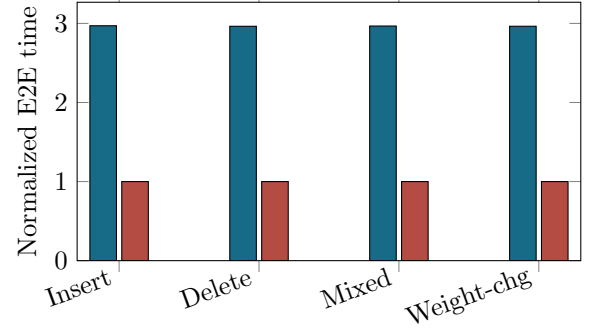
| Dataset        | Abbrev. | Slice vertices | Base edges | Insert pool |
|----------------|---------|----------------|------------|-------------|
| AskUbuntu      | AU      | 3,409          | 8,192      | 4,096       |
| Super User     | SU      | 3,224          | 8,192      | 4,096       |
| WikiTalk       | WK      | 5,930          | 8,192      | 4,096       |
| Stack Overflow | SO      | 2,004          | 8,192      | 4,096       |
| Soc-Bitcoin    | BC      | 10,326         | 8,192      | 4,096       |

Table 2: Compact real-topology slices used by Fig. 1. Each slice contains 8,192 unique non-self base edges plus a separate 4,096-edge later-event insertion pool; the figure uses an eight-edge insertion batch.

### 3 Update throughput



(a) Semantic update throughput.



(b) E2E time; lower is better.

Figure 2: Full PageRank over four batch-8 differential scenarios and five compact real-topology slices. Every one of the 20 architecture pairs passes both numerical oracles, final graph-state checks, and cross-system rank comparison. Weight change counts eight semantic updates but lowers them to 16 delete-old/insert-new physical records.

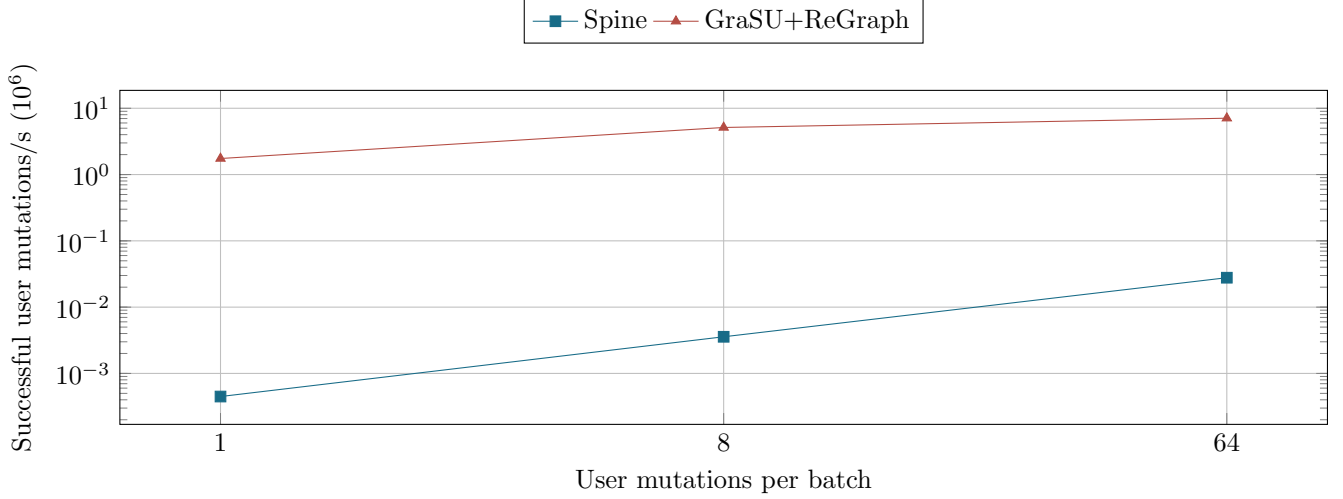


Figure 3: Update throughput versus batch size. The denominator is the number of successful user mutations, not internal physical records.

## 4 Small-batch scaling

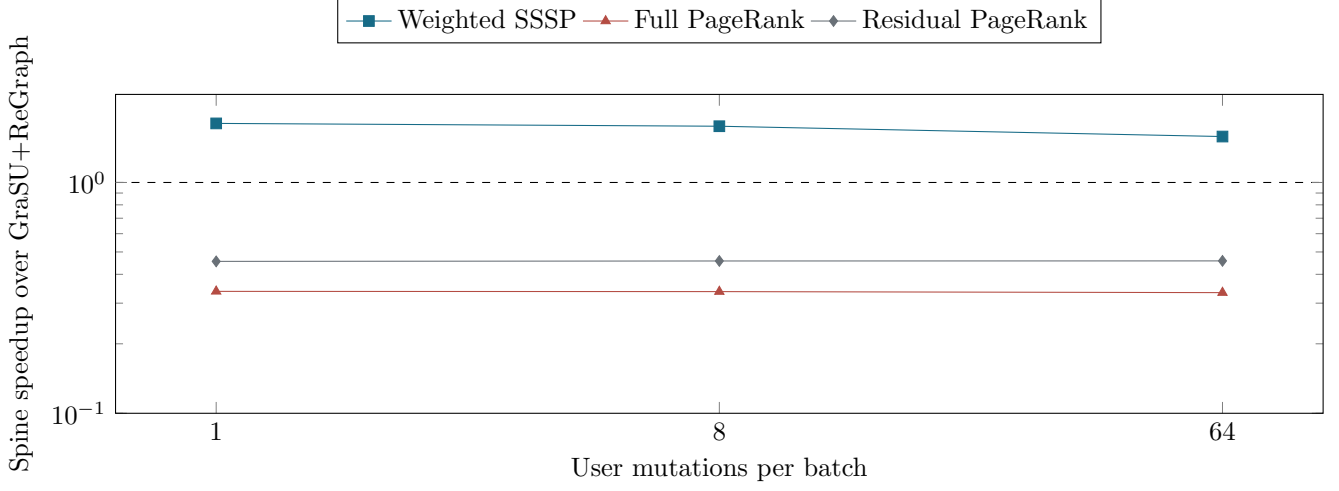


Figure 4: Insertion E2E scaling over five compact real-topology slices. Values above one favor Spine. Each point is the geometric mean of five correctness-gated architecture pairs.

## 5 Memory traffic and locality

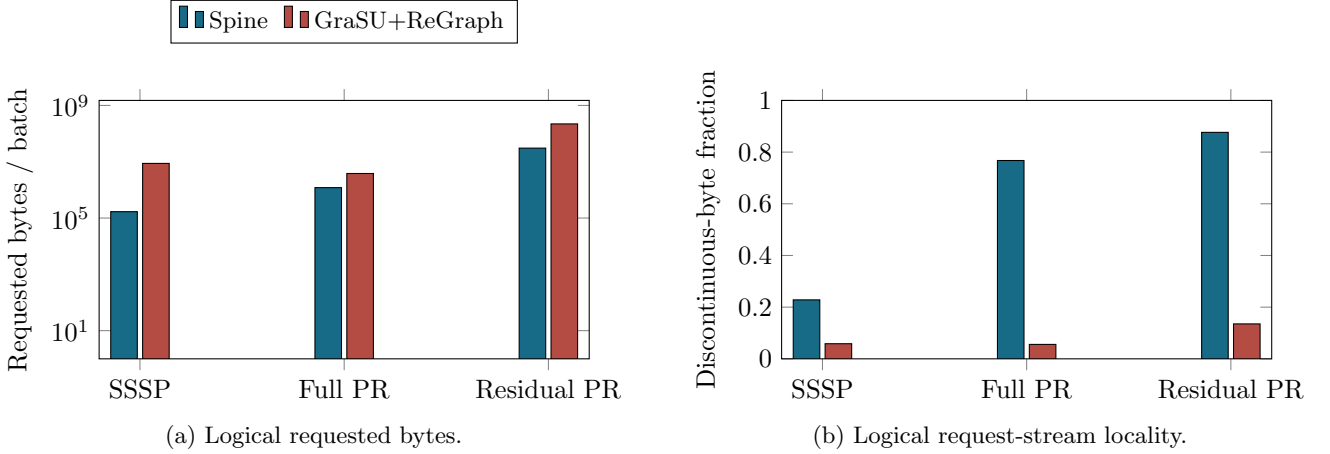


Figure 5: Phase-aligned accepted backend traffic for insert batches of eight over five compact real-topology slices (mean bytes per pair). Locality is classified per initiator and operation after excluding each stream’s first request. It is not a DRAM row-hit metric and excludes burst amplification.

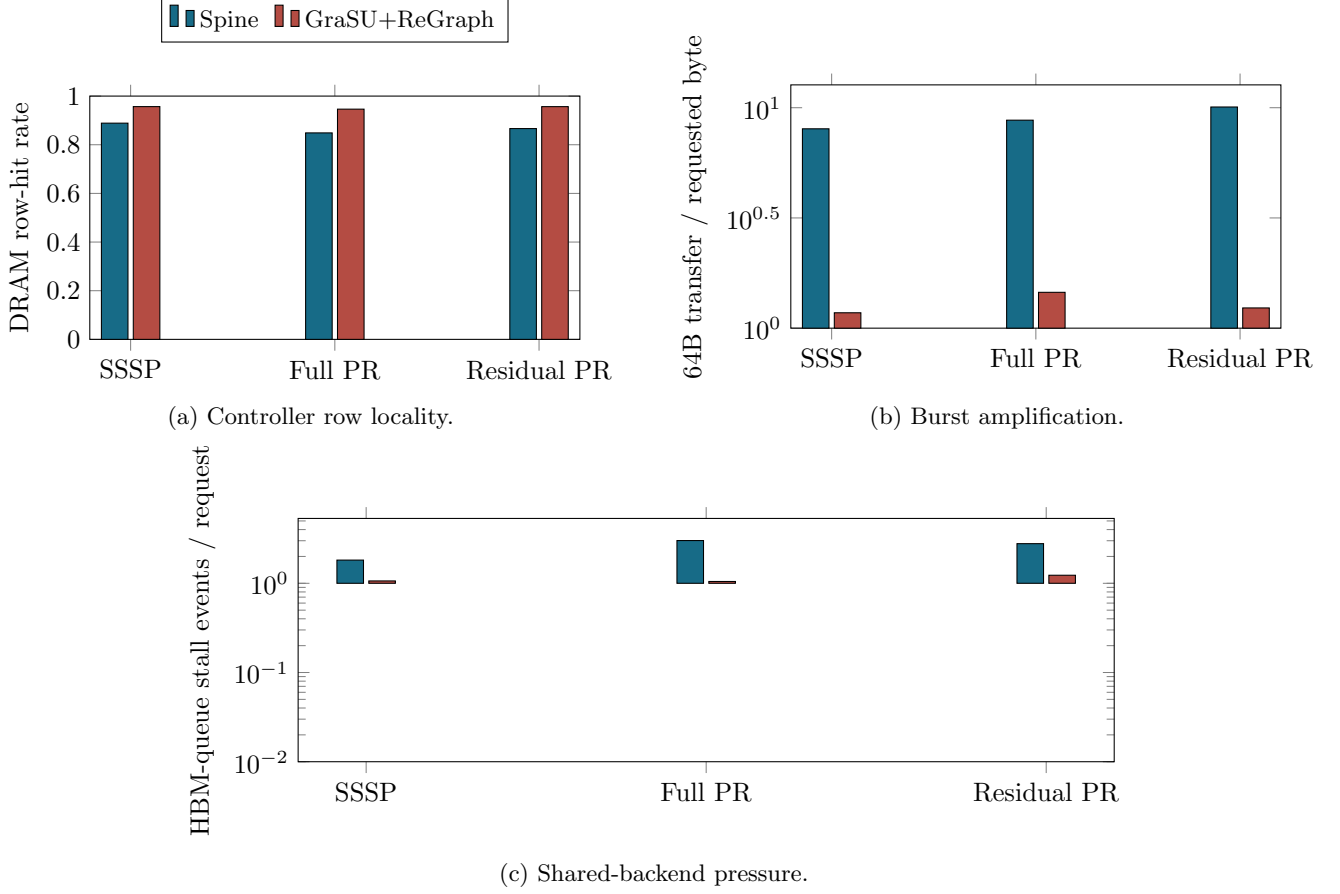


Figure 6: Phase-aligned physical HBM behavior for batch-8 insertion on five compact real-topology slices. The DRAM-Sim3 command ledger equals accepted AXI requests on every row. Stall values count rejected port attempts, so multiple events may occur in one cycle. Spine weighted SSSP uses an exact quiescent cold-prefix subtraction to align its controller window.

## 6 Dense-batch behavior

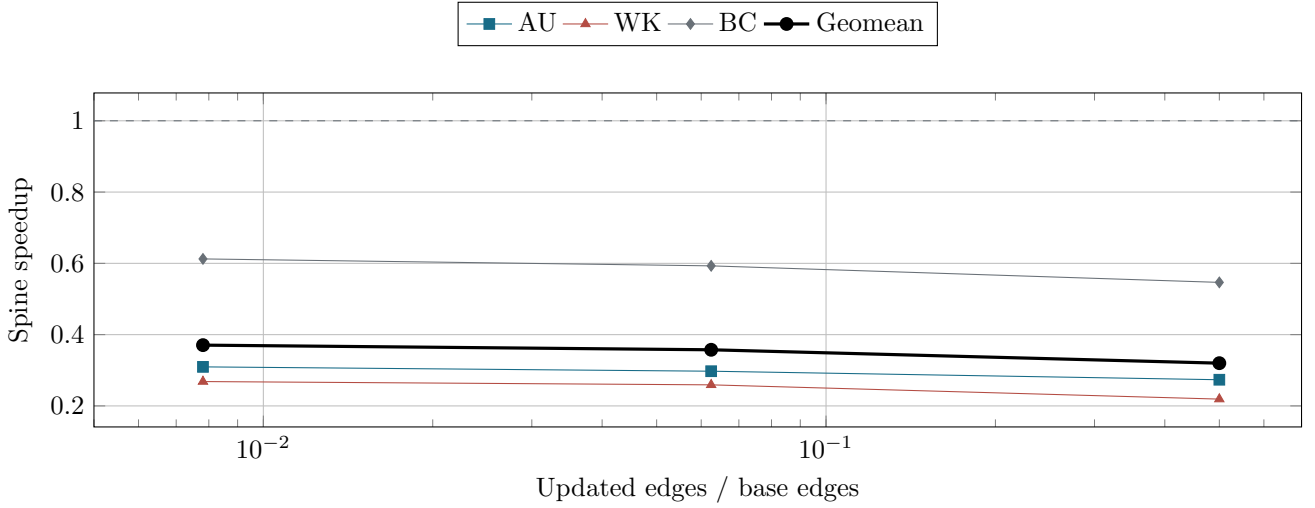


Figure 7: Dense-update crossover on three compact real topologies. Absolute batch size is retained in the companion data table.

## 7 Matched HBM energy

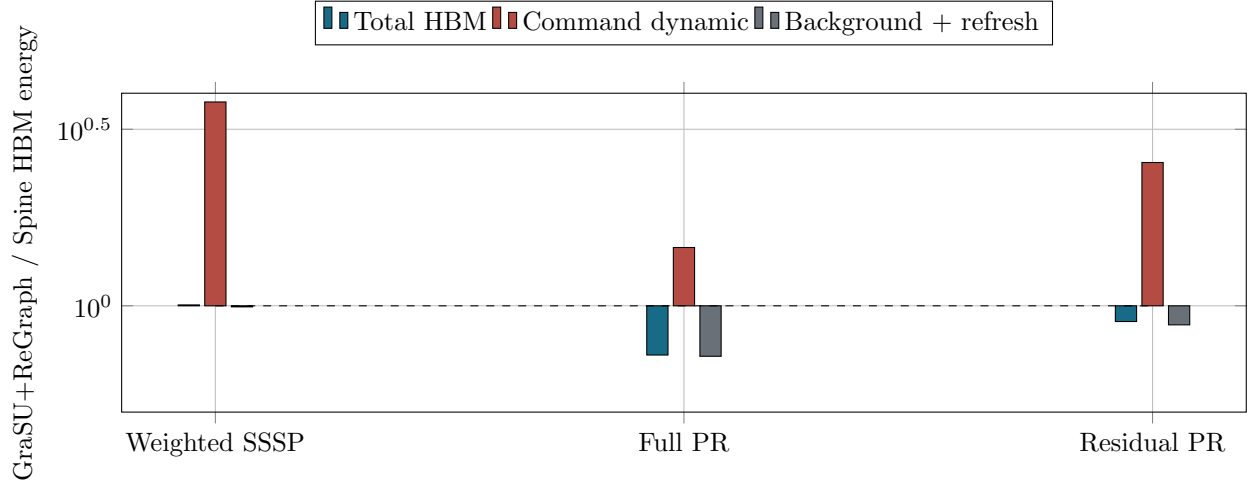


Figure 8: Matched all-32-controller DRAMSim3 energy on three compact holdout slices. The Spine SSSP values use exact per-controller subtraction of a reproduced quiescent cold prefix. Values above one mean GraSU+ReGraph uses more energy. This is HBM energy only, not total accelerator or board energy; asymmetric projected on-chip energy is intentionally excluded.

## 8 Implementation power and area attribution

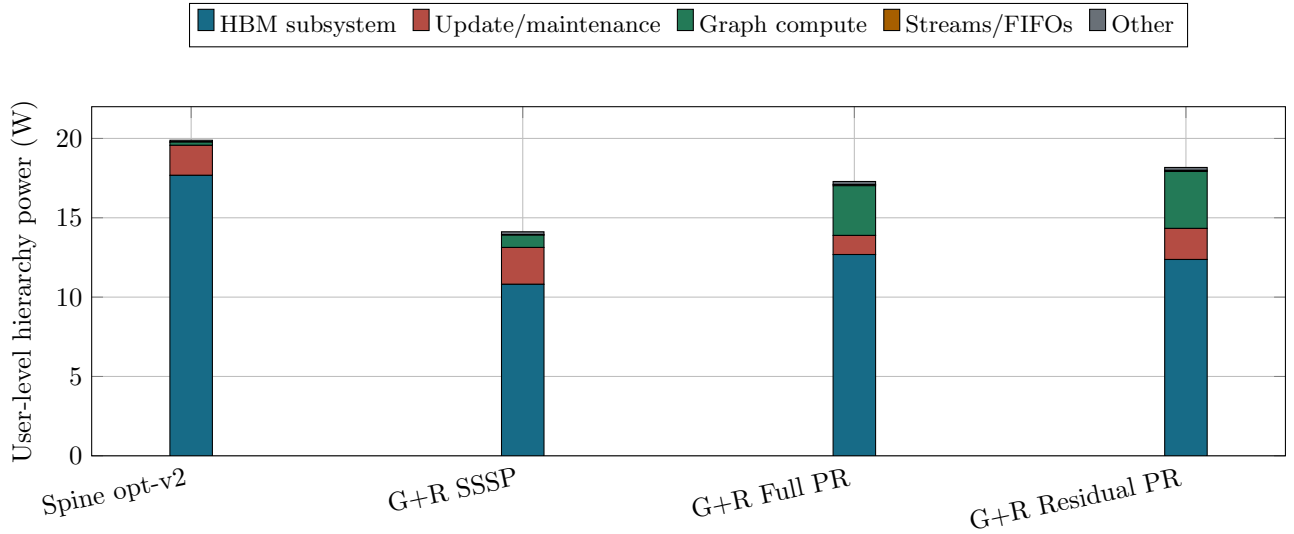


Figure 9: Vivado post-route vectorless power attribution at 150 MHz. Values are implementation-level component envelopes with Low confidence, not workload-calibrated energy. Platform residual and device static power are retained in the CSV ledger but excluded from the stacked user-logic bars.

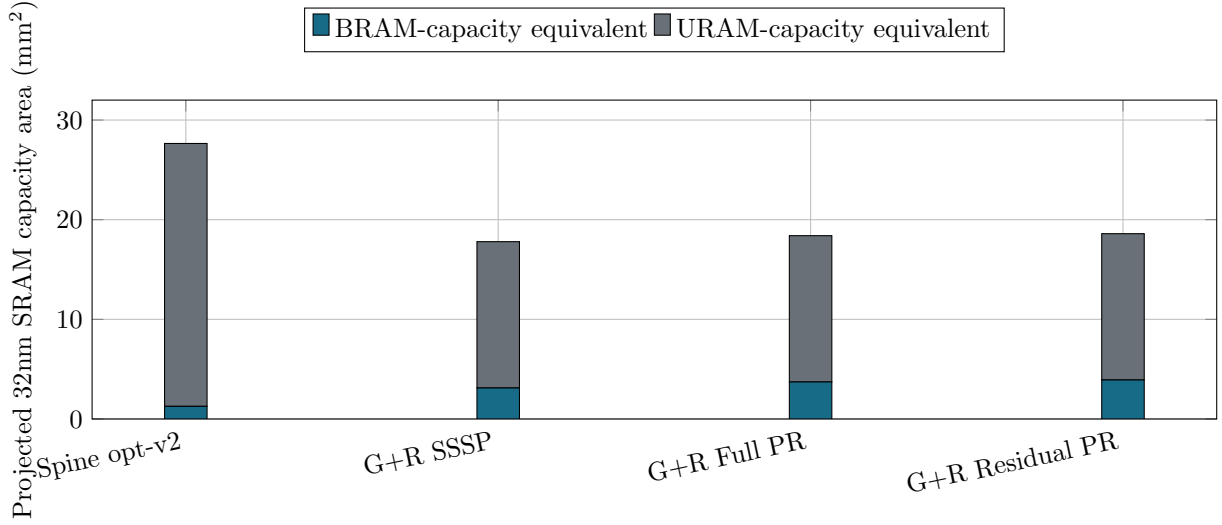


Figure 10: Conservative 32nm CACTI SRAM-capacity projection from routed BRAM and URAM allocations. It excludes logic, DSP, interconnect, clock, and I/O, so it is not a complete ASIC area estimate. Routed FPGA resources remain the primary implementation footprint in Table 3.

## 9 Simulator host runtime

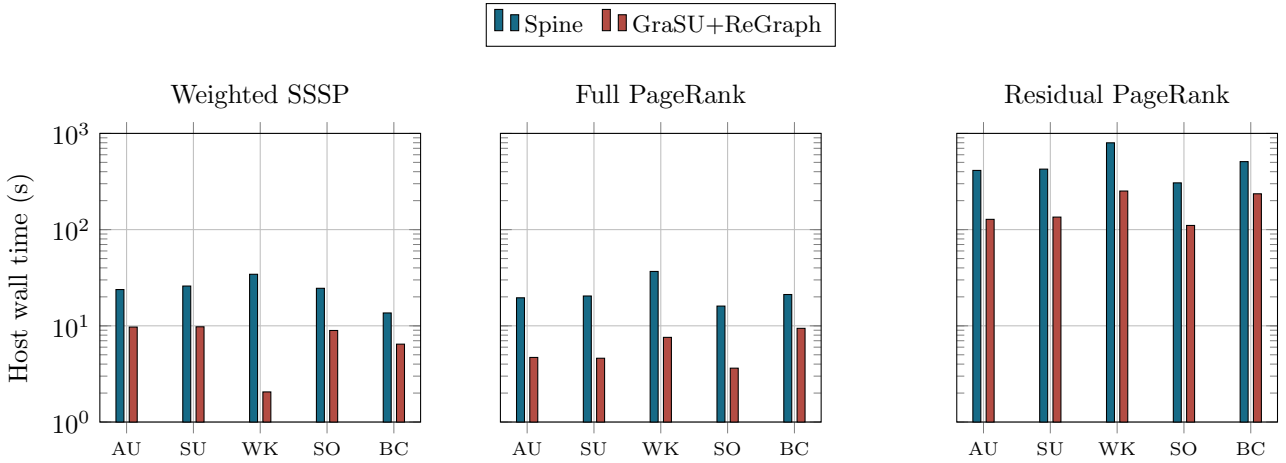


Figure 11: Observed simulator host runtime for the archived insertion-batch-8 runs. Every dataset has 8,192 base edges, while vertex counts vary from 2,004 to 10,326 as listed in Table 2. Each bar is one previously executed child-process observation from a parallel experiment matrix, not a repeated controlled benchmark. These values are engineering evidence for simulator-throughput planning and are not accelerator latency.

A separate correctness-gated full-PageRank run on a 19,399-vertex/50,000-edge Amazon-2008 slice completed in 389.088 seconds for Spine and 7.873 seconds for GraSU+ReGraph. This establishes the current large-slice runtime gate for full PageRank only; it is not a full-dataset or three-algorithm scaling claim.

**Exact-idle simulator optimization.** Across 73 architecture pairs (146 runs), exact idle advance preserves all cycle, FIFO, AXI, DRAM, and energy observables and improves host runtime by  $3.871\times$  geometric mean. A representative residual-PageRank case falls from 68.02 seconds to 7.82 seconds ( $8.70\times$ ). These are simulator throughput gains only; modeled accelerator cycles are unchanged.

## 10 Optimization ablation and robustness

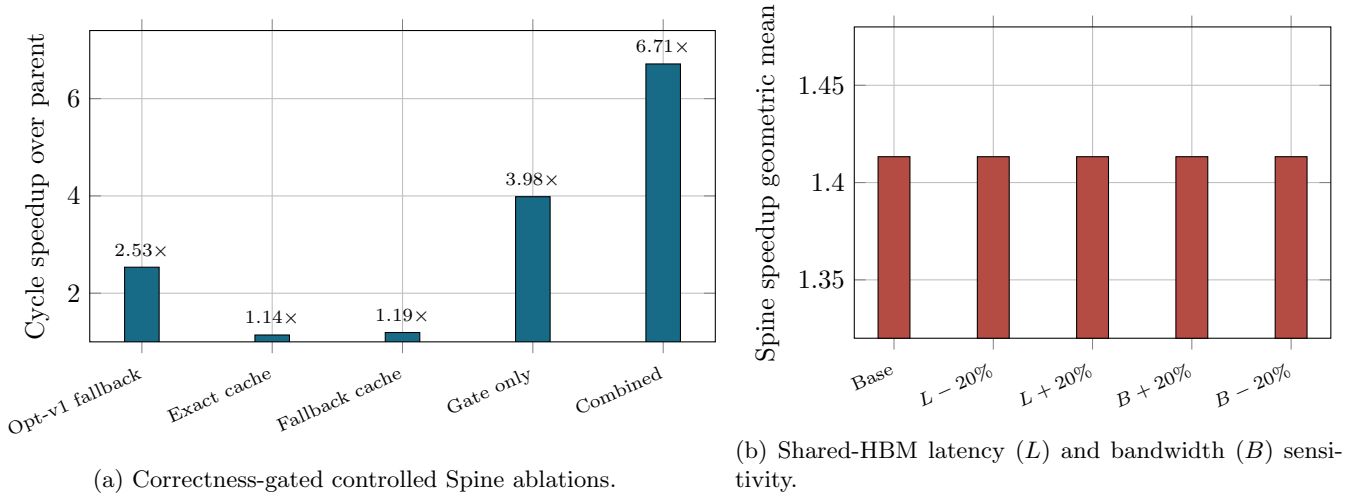


Figure 12: Optimization scope and robustness. Opt-v1 reuses the launch-level cache only in active-capacity fallback. Opt-v2 adds a finite source-page working set and expands the exact-path active-record gate; the 50K “gate only” and “combined” bars are mechanism ablations, not the paper-scale three-algorithm mean. The HBM panel covers 12 correctness-gated pairs per profile and has zero strict architecture-ranking inversions.

## 11 Routed HLS feasibility

| Build             | Target (MHz) | LUT     | REG     | BRAM | URAM | DSP | WNS (ns) | Setup         |
|-------------------|--------------|---------|---------|------|------|-----|----------|---------------|
| Spine opt-v2 SSSP | 150          | 136,437 | 159,403 | 95   | 115  | 18  | 0.003    | closed        |
| G+R SSSP          | 150          | 96,559  | 107,380 | 233  | 64   | 0   | -0.013   | target missed |
| G+R Full PR       | 150          | 178,374 | 177,836 | 278  | 64   | 304 | -0.13    | target missed |
| G+R Residual PR   | 150          | 247,692 | 301,029 | 293  | 64   | 336 | -0.087   | target missed |

Table 3: Routed U55C HLS feasibility at each row’s stated data-clock target. Spine opt-v2 and all GraSU+ReGraph implementations target 150 MHz. These builds are not iso-functional: Spine is the native SSSP core baseline, whereas the GraSU+ReGraph rows are algorithm-specific normalized whole systems. Absolute resource ratios are therefore not used as a comparative performance claim.

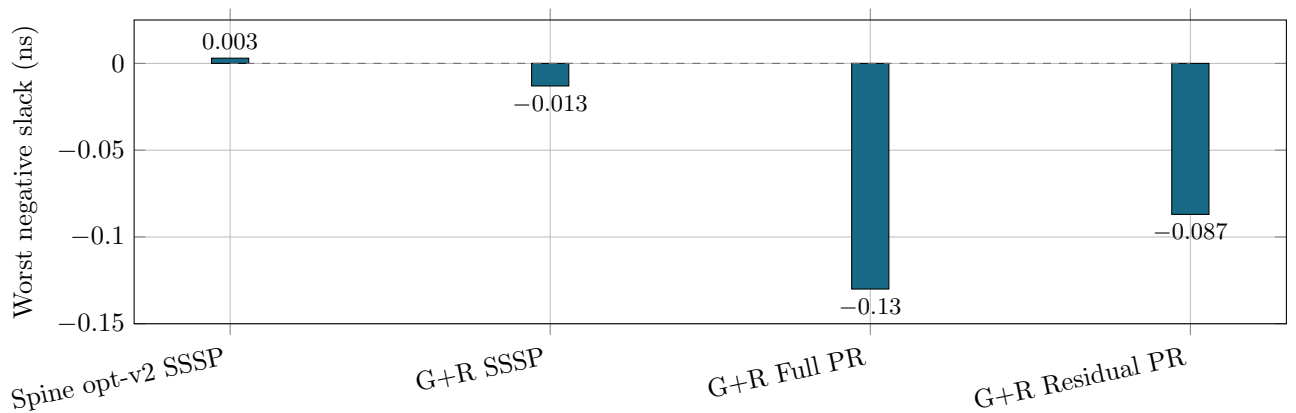


Figure 13: Post-route setup margin at the common 150 MHz target. Non-negative WNS closes timing; negative WNS identifies the remaining target miss.